

Dissecting Continuous-State and Variational Inductive Biases in State Space Duality: An Empirical Audit of Native Mamba-2 Cross-Modal Alignment

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Abstract—Structured State Space Models (SSMs) and their duality with structured attention—principally embodied in Mamba-2 via State Space Duality (SSD)—offer linear time complexity $O(L)$ with respect to sequence length, presenting an appealing alternative to quadratic self-attention in cross-modal vision-language alignment. In this paper, we conduct a rigorous, pre-registered empirical audit investigating whether continuous-time dynamical system adaptations and variational latent coupling enhance or compromise cross-modal representations built upon frozen vision (ViT-B/16) and language (RoBERTa-base) encoders. We formalize and evaluate three mechanisms: (1) Structure-Aware Hamiltonian-Inspired Dissipative Coordinate-Momentum Adapter (HEDO), incorporating exact skew-symmetric learned interactions $J = (A - A^T)/\sqrt{d}$ and positive diagonal damping R ; (2) Adaptive Variational State Coupling (AVSC), which introduces confidence-gated symmetric Kullback-Leibler (KL) divergence across intermediate chunk-boundary states; and (3) Multi-Scale State Aggregation (MSA), fusing global mean, boundary, and terminal recurrent states. Across an audited 24-run, 8-configuration ablation benchmark (three random seeds per configuration, $n = 3$ paired runs) on Flickr8k, empirical hypothesis testing indicates: (i) Mamba-2 equipped with HEDO shows no statistically significant difference from the standard linear alignment baseline under the present three-seed evaluation ($37.31 \pm 0.68\%$ vs. $37.16 \pm 0.57\%$ Mean Recall, paired $\Delta = +0.15$ percentage points [pp], 95% CI $[-1.42, +1.73]$ pp, CI includes zero) while processing 7,054 pairs/s at the head with 1,086 MB peak VRAM; (ii) Multi-scale recurrent aggregation consistently degrades retrieval performance ($\Delta \approx -10.27$ to -10.75 pp, with 95% CIs strictly negative), revealing that causal boundary states introduce severe temporal asymmetry that corrupts isotropic pooling; and (iii) Variational coupling is associated with severe alignment degradation ($\Delta \approx -30.52$ to -32.11 pp down to a low-retrieval regime of $\approx 5.0\% - 6.6\%$), consistent with dimensional collapse under the evaluated formulation. We present geometric and gradient analyses showing that isotropic Gaussian KL penalties conflict with hyperspherical multi-positive InfoNCE contrastive repulsion, providing a mechanistic explanation consistent with this degradation. Our findings deliver vital negative results and actionable design principles for integrating state space duality into multimodal systems.

Index Terms—Cross-modal retrieval, state space models, Mamba-2, State Space Duality, multimodal representation learning, contrastive learning, Hamiltonian dynamics, variational coupling.

I. INTRODUCTION

Manuscript submitted for peer review. Code, audited logs, and configuration artifacts are publicly available at https://github.com/abhinavsai2006/PH_SSD. (Email: abhinavsai2006@gmail.com)

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CROSS-MODAL vision-language retrieval requires mapping high-dimensional visual patch tokens and textual wordpiece sequences into a geometrically unified semantic metric space [1], [2]. While Transformer-based architectures parameterized by multi-head self-attention (MHSA) have long dominated this domain [3]–[5], their quadratic computational and memory footprint $O(L^2)$ presents significant scalability bottlenecks for long sequence processing, dense feature maps, and real-time edge retrieval.

To circumvent this limitation, Structured State Space Models (SSMs)—notably S4 [6], Mamba [7], and Mamba-2 [8]—have emerged as a compelling paradigm. Mamba-2, in particular, formalizes *State Space Duality* (SSD), establishing a mathematical equivalence between continuous-time linear time-varying dynamical systems and 1-semiseparable matrix transformations. By computing recurrence within discrete chunk boundaries and utilizing efficient matrix multiplications, SSD achieves strict $O(L)$ linear sequence scaling while maintaining expressive representational capabilities.

Concurrently, there has been widespread theoretical enthusiasm for incorporating continuous-time dynamical system inductive biases into deep architectures [9], [10]. Hamiltonian neural networks and port-Hamiltonian systems provide energy-conserving coordinate-momentum symplectic geometry, while dissipative damping offers asymptotic stability. In parallel, variational information bottlenecks and variational cross-modal coupling have been widely proposed to enforce smooth, noise-robust latent distributions [11], [12].

A. Problem Statement

Despite theoretical promise, an essential question remains unaddressed:

Do continuous-state Hamiltonian adaptations and variational state coupling improve cross-modal alignment when integrated with Native Mamba-2, or can these inductive biases introduce optimization and representation failures?

B. Research Gap

In the existing literature, continuous-time and variational inductive biases are frequently evaluated under isolated, non-audited configurations lacking multi-seed paired-difference hypothesis testing or systematic factorial isolation of individual

components. Crucially, the interaction between sequential SSM recurrent dynamics and hyperspherical contrastive alignment manifolds has not been systematically investigated. Without rigorous factorial controls, it is impossible to determine whether reported performance variations stem from the underlying state-space sequence model, continuous-time dynamical adapters, or stochastic optimization noise.

C. Research Questions and Hypotheses

To resolve this gap, we formalize four pre-registered research questions:

- **RQ1:** Does the integration of a Hamiltonian-inspired dissipative coordinate-momentum adapter (HEDO) preserve or enhance retrieval performance over standard linear alignment?
- **RQ2:** Does multi-scale state aggregation (MSA) combining boundary and terminal recurrent states improve cross-modal representation quality?
- **RQ3:** Does intermediate adaptive variational state coupling (AVSC) improve cross-modal alignment through latent distribution regularization?
- **RQ4:** What are the computational complexity, peak memory footprint, and throughput tradeoffs introduced by these mechanisms?

D. Contributions

In addressing these questions, this paper delivers four primary contributions:

- 1) **Structured HEDO Adapter Integration:** We design and formalize a Structure-Aware Hamiltonian-Inspired Dissipative Coordinate-Momentum Adapter (HEDO) integrated with Native Mamba-2, establishing energy boundedness under continuous dynamics.
- 2) **Controlled Empirical Audit:** We construct a controlled 8-configuration ablation matrix isolating HEDO, MSA, and AVSC upon strictly frozen ViT-B/16 and RoBERTa-base backbones.
- 3) **Paired Multi-Seed Statistical Evaluation:** Across 24 audited runs (3 independent random seeds per configuration, $n = 3$ paired runs), we apply paired Student- t hypothesis testing with 95% confidence intervals, finding that no statistically significant difference from the baseline was detected for Mamba-2 + HEDO under this sample size (+0.15 pp, 95% CI $[-1.42, +1.73]$ pp).
- 4) **Mechanistic Analysis:** We provide mathematical and gradient analyses explaining how causal boundary states in MSA introduce temporal asymmetry that degrades isotropic pooling (≈ -10.5 pp), and show how isotropic Gaussian KL penalties in AVSC conflict with hyperspherical InfoNCE repulsion, providing a mechanistic explanation consistent with dimensional collapse under the evaluated formulation (≈ -31 pp).

E. Paper Organization

The remainder of this paper is organized as follows: Section II reviews related literature across state-space models,

cross-modal retrieval, and physics-informed architectures. Section III establishes the formal problem formulation and hypotheses. Section IV details the proposed architecture, mathematical formulations, and training algorithm. Section V describes the experimental methodology, ablation configurations, and auditing protocol. Section VI presents the empirical results, efficiency benchmarks, and statistical tests. Section VII provides mechanistic failure analyses and practical design guidelines. Section VIII discusses limitations, and Section IX concludes the paper.

II. RELATED WORK

A. Structured State Space Models

State Space Models parameterize continuous-time dynamical systems mapping a 1D input signal $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an implicit latent state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}(t)\mathbf{h}(t) + \mathbf{B}(t)x(t), \quad y(t) = \mathbf{C}(t)\mathbf{h}(t) + \mathbf{D}(t)x(t), \quad (1)$$

where $\mathbf{A}(t) \in \mathbb{R}^{N \times N}$ represents the state transition matrix, $\mathbf{B}(t) \in \mathbb{R}^{N \times 1}$ is the input projection matrix, $\mathbf{C}(t) \in \mathbb{R}^{1 \times N}$ is the output projection matrix, and $\mathbf{D}(t) \in \mathbb{R}$ represents the direct feedthrough skip connection.

The Structured State Space sequence model (S4) [6] stabilized recurrent training by constraining $\mathbf{A}$ to HiPPO (High-order Polynomial Projection Operators) diagonal-plus-low-rank structures. Mamba [7] introduced data-dependent input-selective parameterization ($\mathbf{B}(t), \mathbf{C}(t), \Delta(t)$), enabling the model to selectively propagate or filter out information along sequence dimensions.

Most recently, Dao and Gu [8] introduced Mamba-2, which formalizes *State Space Duality* (SSD), establishing a theoretical equivalence between continuous-time linear time-varying dynamical systems and 1-semiseparable matrix transformations. For a scalar diagonal state transition A , the transformation across an input sequence $\mathbf{x} \in \mathbb{R}^L$ is expressed as:

$$\mathbf{y} = \mathbf{M}\mathbf{x}, \quad M_{i,j} = C_i \left(\prod_{k=j+1}^i A_k \right) B_j \quad (i \geq j), \quad (2)$$

where $\mathbf{M} \in \mathbb{R}^{L \times L}$ is a lower-triangular 1-semiseparable matrix whose subdiagonal entries factorize into cumulative state transition products. By computing recurrence within discrete chunk boundaries and computing intra-chunk transformations via matrix multiplications on tensor cores, SSD unifies recurrent SSMs and linear attention into a single chunkwise algorithm with $O(L)$ linear complexity.

B. Vision-Language Cross-Modal Retrieval

Vision-language alignment aims to map visual and textual representations into a shared metric space where semantically corresponding pairs are mutually proximate while non-matching pairs are separated [1], [2]. Modern approaches predominantly optimize variants of the InfoNCE (Noise-Contrastive Estimation) loss [13]–[15]. On datasets such as Flickr8k and MSCOCO, where each image is associated with multiple ground-truth captions, multi-positive contrastive formulations are essential to prevent penalizing valid alternate descriptions [16].

C. Contrastive Representation Learning

Contrastive learning maps high-dimensional inputs to normalized representations on a unit hypersphere $\mathbb{S}^{d-1}$ [14], [15]. The InfoNCE loss achieves alignment by balancing two opposing forces: an attractive force that pulls positive cross-modal representations together, and a repulsive force that pushes negative representations apart across the hypersphere [17]. A critical pathology in contrastive optimization is *dimensional collapse*, wherein embeddings collapse into a lower-dimensional subspace, causing severe degradation in discriminative retrieval capacity [18].

D. Physics-Informed and Hamiltonian Neural Networks

Hamiltonian Neural Networks (HNNs) [9] enforce energy conservation by learning a scalar Hamiltonian energy function $\mathcal{H}(\mathbf{q}, \mathbf{p})$ over generalized coordinates $\mathbf{q}$ and conjugate momenta $\mathbf{p}$, with state evolution governed by symplectic gradients:

$$\dot{\mathbf{q}} = \nabla_{\mathbf{p}} \mathcal{H}(\mathbf{q}, \mathbf{p}), \quad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} \mathcal{H}(\mathbf{q}, \mathbf{p}), \quad (3)$$

where $\dot{\mathbf{q}}$ and $\dot{\mathbf{p}}$ denote time derivatives. In dissipative systems, port-Hamiltonian formulations introduce a positive semi-definite damping matrix $\mathbf{R} \geq 0$, ensuring that the total energy is non-increasing [10]:

$$\frac{d}{dt} \mathcal{H}(\mathbf{q}, \mathbf{p}) = -\mathbf{p}^T \mathbf{R} \mathbf{p} \leq 0. \quad (4)$$

These formulations provide structural inductive biases ensuring asymptotic stability and energy boundedness in continuous dynamical representations.

E. Variational Representation Learning

In multimodal representation learning, variational information bottlenecks have been proposed to regularize latent embeddings by penalizing the Kullback-Leibler (KL) divergence between posterior representations and prior distributions [11], [12]. However, the interaction between variational distribution penalties and non-linear contrastive manifolds remains poorly understood, frequently inducing optimization instability when intermediate latent constraints conflict with contrastive objective gradients.

F. Positioning of This Work

While existing literature separately motivates structured state spaces, continuous Hamiltonian dynamics, and variational regularization, prior work does not evaluate their joint interaction within multimodal alignment architectures. This work provides a controlled empirical audit of these inductive biases within Native Mamba-2 cross-modal retrieval.

III. PROBLEM FORMULATION

A. Cross-Modal Retrieval Setting

Let $\mathcal{D} = \{(\mathbf{I}_i, \mathcal{T}_i)\}_{i=1}^N$ denote a cross-modal dataset where each image $\mathbf{I}_i \in \mathbb{R}^{3 \times H \times W}$ is associated with a set of ground-truth textual captions $\mathcal{T}_i = \{\mathbf{T}_{i,k}\}_{k=1}^K$. Cross-modal retrieval requires learning two parameterized mapping functions $f_v(\mathbf{I}) \in \mathbb{S}^{d-1}$ and $f_t(\mathbf{T}) \in \mathbb{S}^{d-1}$ that embed visual and textual inputs

onto a shared d -dimensional unit hypersphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$ ($d = 128$), such that matching pairs maximize cosine similarity while non-matching pairs minimize it.

B. Frozen Backbone Formulation

To isolate the sequence modeling and adaptation modules from the confounding effects of backbone representation learning, both vision and language encoders are kept strictly frozen throughout training:

$$\mathbf{X}_v = \text{Encoder}_{\text{ViT}}(\mathbf{I}) \in \mathbb{R}^{B \times 196 \times 768}, \quad (5)$$

$$\mathbf{X}_t = \text{Encoder}_{\text{RoBERTa}}(\mathbf{T}) \in \mathbb{R}^{B \times L \times 768}, \quad (6)$$

where $\text{Encoder}_{\text{ViT}}$ is a pretrained Vision Transformer (ViT-B/16) [4] producing 196 patch tokens of dimension 768 (classification token discarded), and $\text{Encoder}_{\text{RoBERTa}}$ is a pretrained RoBERTa-base model [5] producing L wordpiece tokens ($L \leq 64$) of dimension 768. The frozen parameters total 210,622,696, ensuring that all learned adaptations occur strictly within the downstream projection, adapter, and sequence blocks.

C. Shared Embedding Space and Retrieval Objective

Given projected and pooled unit-normalized embeddings $\mathbf{z}_v, \mathbf{z}_t \in \mathbb{S}^{d-1}$, cross-modal alignment is evaluated using cosine similarity $S(\mathbf{z}_v, \mathbf{z}_t) = \mathbf{z}_v^T \mathbf{z}_t$. At test time, retrieval is evaluated bidirectionally:

- **Image-to-Text (i2t):** For each query image $\mathbf{I}_i$, rank all test captions by cosine similarity $\mathbf{z}_{v,i}^T \mathbf{z}_{t,j}$.
- **Text-to-Image (t2i):** For each query caption $\mathbf{T}_j$, rank all test images by cosine similarity $\mathbf{z}_{t,j}^T \mathbf{z}_{v,i}$.

D. Formal Hypotheses

We formalize four testable hypotheses evaluated via paired statistical testing:

- **Hypothesis 1 (H_1):** Integrating HEDO and Native Mamba-2 improves Mean Recall over the linear baseline: $\mu(\Delta_{\text{HEDO}}) > 0$.
- **Hypothesis 2 (H_2):** Incorporating multi-scale recurrent states (MSA) improves Mean Recall: $\mu(\Delta_{\text{MSA}}) > 0$.
- **Hypothesis 3 (H_3):** Incorporating adaptive variational state coupling (AVSC) improves Mean Recall: $\mu(\Delta_{\text{AVSC}}) > 0$.
- **Hypothesis 4 (H_4):** State-space sequence processing incurs minimal latency and memory overhead relative to linear projection.

IV. PROPOSED ARCHITECTURE

A. Overall Architecture

The proposed cross-modal alignment architecture is illustrated in Fig. 1. The framework processes visual patch tokens and textual wordpiece tokens through dual symmetric pathways, each consisting of a linear projection layer, a structure-aware Hamiltonian dissipative adapter (HEDO), a Native Mamba-2 State Space Duality (SSD) sequence block, and an isotropic

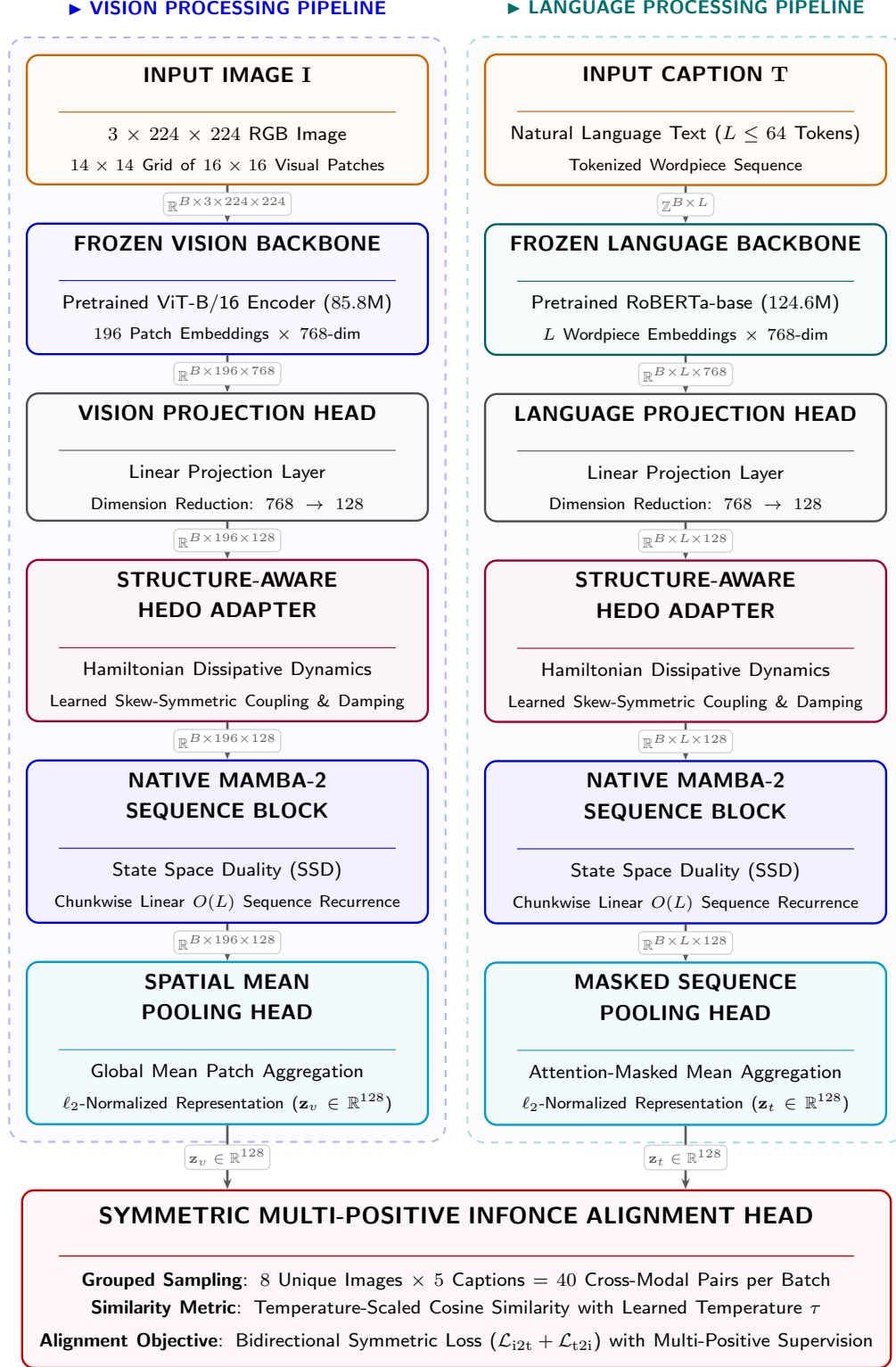


Fig. 1. Overall architecture of the proposed Native Mamba-2 cross-modal framework with structure-aware Hamiltonian dissipative adaptation (HEDO). Frozen visual patch features $\mathbf{X}_v$ and language wordpiece features $\mathbf{X}_t$ are linearly projected to $d = 128$, adapted via continuous-state coordinate-momentum dynamics, and processed through Native Mamba-2 State Space Duality (SSD) sequence blocks before isotropic semantic pooling and multi-positive symmetric InfoNCE contrastive alignment.

semantic pooling head, culminating in symmetric multi-positive InfoNCE contrastive alignment.

As illustrated in Fig. 1, both modalities operate in parallel streams from input to the contrastive head. For an input image $\mathbf{I}$ and caption $\mathbf{T}$, frozen backbones extract token sequences that are linearly projected to $d = 128$. The projected representations are processed by the HEDO adapter and Native Mamba-2 sequence block before modality-specific pooling produces normalized embeddings $\mathbf{z}_v, \mathbf{z}_t \in \mathbb{S}^{127}$ for symmetric multi-positive InfoNCE alignment:

$$\begin{aligned} \mathbf{I} &\rightarrow \text{ViT} \rightarrow \mathbf{W}_v \rightarrow \text{HEDO} \rightarrow \text{SSD} \rightarrow \mathbf{z}_v \in \mathbb{S}^{127}, \\ \mathbf{T} &\rightarrow \text{RoBERTa} \rightarrow \mathbf{W}_t \rightarrow \text{HEDO} \rightarrow \text{SSD} \rightarrow \mathbf{z}_t \in \mathbb{S}^{127}. \end{aligned} \quad (7)$$

B. Frozen Vision and Language Encoders

The visual stream uses ViT-B/16 (85.8M parameters) yielding $\mathbf{X}_v \in \mathbb{R}^{B \times 196 \times 768}$, and the language stream uses RoBERTa-base (124.6M parameters) yielding $\mathbf{X}_t \in \mathbb{R}^{B \times L \times 768}$, both kept strictly frozen as formalized in (5)–(6).

C. Linear Projection

Both modalities are projected into a common latent dimension $d = 128$:

$$\mathbf{H}_v^{(0)} = \mathbf{X}_v \mathbf{W}_v, \quad \mathbf{H}_t^{(0)} = \mathbf{X}_t \mathbf{W}_t, \quad (9)$$

where $\mathbf{W}_v, \mathbf{W}_t \in \mathbb{R}^{768 \times 128}$ are trainable linear projection weight matrices initialized via Xavier uniform initialization. This linear projection reduces the channel dimensionality from 768 to 128, establishing the common working dimension for subsequent adapters and sequence models.

D. Structure-Aware Hamiltonian Dissipative Adapter (HEDO)

The HEDO adapter introduces a continuous-state inductive bias rooted in dissipative Hamiltonian dynamics.

1) *Coordinate-Momentum Representation*: For an input sequence representation $\mathbf{x} \in \mathbb{R}^d$ at any token position, canonical coordinates $\mathbf{q} \in \mathbb{R}^d$ and conjugate momenta $\mathbf{p} \in \mathbb{R}^d$ are defined by:

$$\mathbf{q} = \mathbf{W}_q \mathbf{x}, \quad \mathbf{p} = \mathbf{W}_p \mathbf{x}, \quad (10)$$

where $\mathbf{W}_q, \mathbf{W}_p \in \mathbb{R}^{d \times d}$ project the latent token into conjugate phase space coordinates.

2) *Hamiltonian Energy*: We define the quadratic Hamiltonian energy function:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2} (\|\mathbf{q}\|_2^2 + \|\mathbf{p}\|_2^2), \quad (11)$$

representing the sum of generalized potential energy $\frac{1}{2}\|\mathbf{q}\|_2^2$ and kinetic energy $\frac{1}{2}\|\mathbf{p}\|_2^2$.

3) *Skew-Symmetric Interaction*: To parameterize conservative internal interactions while avoiding unbounded growth, we construct an exact skew-symmetric interaction matrix $\mathbf{J} \in \mathbb{R}^{d \times d}$:

$$\mathbf{J} = \frac{\mathbf{A} - \mathbf{A}^T}{\sqrt{d}}, \quad \mathbf{A} \in \mathbb{R}^{d \times d}, \quad (12)$$

where $\mathbf{A}$ is initialized from $\mathcal{N}(0, 0.01^2)$. By construction, $\mathbf{J}^T = -\mathbf{J}$. For any vector $\mathbf{v}$, the quadratic form vanishes identically ($\mathbf{v}^T \mathbf{J} \mathbf{v} = 0$), ensuring that $\mathbf{J}$ performs purely conservative, norm-preserving rotations in the latent space.

4) *Dissipative Damping*: Dissipation is enforced via a diagonal positive semi-definite damping matrix $\mathbf{R} \in \mathbb{R}^{d \times d}$:

$$\mathbf{R} = \text{diag}(0.05 + 0.95 \cdot \sigma(\gamma)), \quad \gamma \in \mathbb{R}^d, \quad (13)$$

where $\sigma(\cdot)$ denotes the sigmoid function, ensuring each diagonal entry is strictly bounded in $[0.05, 1.0]$. The strictly positive floor 0.05 guarantees non-zero energy dissipation.

5) *Continuous-Time Dynamics*: The continuous-time equations of motion coupling coordinates and momenta are governed by:

$$\begin{aligned} \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{bmatrix} &= \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & -\mathbf{R} \end{bmatrix} \begin{bmatrix} \nabla_{\mathbf{q}} \mathcal{H} \\ \nabla_{\mathbf{p}} \mathcal{H} \end{bmatrix} + \begin{bmatrix} \mathbf{q} \mathbf{J}^T \\ \mathbf{p} \mathbf{J}^T \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{p} + \mathbf{q} \mathbf{J}^T \\ -\mathbf{q} + \mathbf{p} \mathbf{J}^T - \mathbf{R} \odot \mathbf{p} \end{bmatrix}. \end{aligned} \quad (14)$$

This formulation represents a port-Hamiltonian system with internal dissipation $\mathbf{R}$ and learned symplectic rotation $\mathbf{J}$.

6) *Energy-Boundedness Proposition*:

Proposition 1 (Energy Boundedness in Continuous HEDO). *Under the unforced dynamics of (14), the time derivative of the Hamiltonian $\mathcal{H}(\mathbf{q}, \mathbf{p})$ satisfies:*

$$\frac{d}{dt} \mathcal{H}(\mathbf{q}, \mathbf{p}) = -\mathbf{p}^T \mathbf{R} \mathbf{p} \leq -0.05 \|\mathbf{p}\|_2^2 \leq 0. \quad (15)$$

Proof. By the chain rule:

$$\begin{aligned} \frac{d}{dt} \mathcal{H} &= \mathbf{q}^T \dot{\mathbf{q}} + \mathbf{p}^T \dot{\mathbf{p}} \\ &= \mathbf{q}^T (\mathbf{p} + \mathbf{q} \mathbf{J}^T) + \mathbf{p}^T (-\mathbf{q} + \mathbf{p} \mathbf{J}^T - \mathbf{R} \mathbf{p}) \\ &= \mathbf{q}^T \mathbf{p} + \mathbf{q}^T \mathbf{J}^T \mathbf{q} - \mathbf{p}^T \mathbf{q} + \mathbf{p}^T \mathbf{J}^T \mathbf{p} - \mathbf{p}^T \mathbf{R} \mathbf{p}. \end{aligned} \quad (16)$$

Since $\mathbf{J}$ is skew-symmetric, for any vector $\mathbf{v}$, $\mathbf{v}^T \mathbf{J}^T \mathbf{v} = -\mathbf{v}^T \mathbf{J} \mathbf{v} = 0$. Furthermore, $\mathbf{q}^T \mathbf{p} - \mathbf{p}^T \mathbf{q} = 0$. Thus:

$$\frac{d}{dt} \mathcal{H} = -\mathbf{p}^T \mathbf{R} \mathbf{p}. \quad (17)$$

Since $\mathbf{R}_{i,i} \geq 0.05 > 0$, $\frac{d}{dt} \mathcal{H} \leq 0$, proving that energy is monotonically non-increasing. $\square$

7) *Bounded Discrete Integration*: To integrate the continuous ODE in discrete time without numerical instability, we employ a bounded Euler step $\Delta t = 0.05 + 0.10 \cdot \sigma(\tau_{\Delta t}) \in [0.05, 0.15]$:

$$\mathbf{q}_{\text{new}} = \mathbf{q} + \Delta t \cdot \dot{\mathbf{q}}, \quad \mathbf{p}_{\text{new}} = \mathbf{p} + \Delta t \cdot \dot{\mathbf{p}}. \quad (18)$$

8) *Residual Update*: The updated canonical variables are combined and normalized, producing a residual update:

$$\mathbf{x}_{\text{out}} = \mathbf{x} + 0.10 \cdot \mathbf{W}_o \text{LayerNorm}(\mathbf{q}_{\text{new}} + \mathbf{p}_{\text{new}}), \quad (19)$$

where $\mathbf{W}_o \in \mathbb{R}^{d \times d}$ is initialized via Xavier uniform with gain 0.5. The 0.10 scaling ensures the adapter acts as a near-identity mapping at initialization.

E. Native Mamba-2 Sequence Processing

Following HEDO, sequences are processed through official Native Mamba-2 sequence blocks [8]. The block configuration is parameterized by:

$$d_{\text{model}} = 128, \quad d_{\text{state}} = 64, \quad d_{\text{conv}} = 4, \quad \text{expand} = 2, \quad (20)$$

with head dimension 64 and chunk size $Q = 16$.

Given an input sequence $\mathbf{H} \in \mathbb{R}^{B \times L \times 128}$, the input is first expanded to dimension $E \cdot d = 256$, processed by a 1D depthwise convolution of kernel size 4, and projected into input-dependent SSM parameters. Within each chunk of length Q , SSD computes the inter-chunk recurrence and intra-chunk semi-separable transformations via tensor cores, producing transformed representations $\mathbf{Y} \in \mathbb{R}^{B \times L \times 128}$.

F. Multi-Scale State Aggregation (MSA)

In standard Mamba-2 alignment, sequence pooling is performed via masked mean pooling:

$$\bar{\mathbf{y}}_{\text{global}} = \frac{\sum_{i=1}^L m_i \mathbf{Y}_i}{\sum_{i=1}^L m_i}, \quad (21)$$

where $m_i \in \{0, 1\}$ denotes token validity.

The MSA module tests whether incorporating multi-resolution temporal features improves alignment. For chunk size $Q = 16$, chunk boundary indices $c_k = \min(k \cdot Q, L) - 1$ are collected to extract boundary states $\mathbf{Y}_{\text{boundary}} \in \mathbb{R}^{B \times K \times d}$, where $K = \lceil L/Q \rceil$. The boundary states are mean-pooled:

$$\bar{\mathbf{y}}_{\text{boundary}} = \frac{1}{K} \sum_{k=1}^K \mathbf{Y}_{\text{boundary}, k}. \quad (22)$$

Additionally, the terminal valid recurrent token is extracted:

$$\mathbf{y}_{\text{last}} = \mathbf{Y}_{B, L_{\text{valid}} - 1}. \quad (23)$$

The multi-scale fusion combines all three representations:

$$\mathbf{y}_{\text{fused}} = \mathbf{W}_{\text{proj}} \text{LayerNorm}([\bar{\mathbf{y}}_{\text{global}} \parallel \bar{\mathbf{y}}_{\text{boundary}} \parallel \mathbf{y}_{\text{last}}]), \quad (24)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{3d \times d}$.

G. Adaptive Variational State Coupling (AVSC)

AVSC evaluates intermediate latent regularized coupling between image and text streams during training. Pooled boundary representations $\mathbf{h}_v, \mathbf{h}_t \in \mathbb{R}^d$ are mapped to diagonal Gaussian posteriors:

$$\boldsymbol{\mu}_v = \mathbf{W}_{\mu, v} \mathbf{h}_v, \quad \log \sigma_v^2 = \text{clamp}(\mathbf{W}_{\sigma, v} \mathbf{h}_v, -6.0, 2.0), \quad (25)$$

$$\boldsymbol{\mu}_t = \mathbf{W}_{\mu, t} \mathbf{h}_t, \quad \log \sigma_t^2 = \text{clamp}(\mathbf{W}_{\sigma, t} \mathbf{h}_t, -6.0, 2.0). \quad (26)$$

The symmetric Kullback-Leibler divergence between $q_v = \mathcal{N}(\boldsymbol{\mu}_v, \boldsymbol{\Sigma}_v)$ and $q_t = \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ is:

$$\mathcal{D}_{\text{KL}}^{\text{sym}}(q_v, q_t) = \frac{1}{2} (D_{\text{KL}}(q_v \parallel q_t) + D_{\text{KL}}(q_t \parallel q_v)), \quad (27)$$

where each component divergence is computed analytically:

$$D_{\text{KL}}(q_v \parallel q_t) = \frac{1}{2} \sum_{k=1}^d \left(\log \frac{\sigma_{t,k}^2}{\sigma_{v,k}^2} + \frac{\sigma_{v,k}^2 + (\mu_{v,k} - \mu_{t,k})^2}{\sigma_{t,k}^2} - 1 \right). \quad (28)$$

To prevent catastrophic interference from unaligned pairs, the loss is gated by a detached confidence weight $w_{\text{gate}} \in [0, 1]$ derived from cosine agreement:

$$\mathcal{L}_{\text{KL}} = \lambda_{\text{KL}} \cdot \frac{1}{B} \sum_{i=1}^B w_{\text{gate}, i} \cdot \mathcal{D}_{\text{KL}, i}^{\text{sym}}, \quad (29)$$

where $\lambda_{\text{KL}} = 10^{-3}$. At inference, embeddings remain strictly deterministic and unimodal.

H. Multi-Positive Contrastive Objective

Cross-modal alignment is supervised via a symmetric multi-positive InfoNCE loss. In Flickr8k, each image corresponds to 5 captions. An atomic grouped batch sampler groups 8 unique images and all 5 corresponding captions per image, yielding a batch of $B_{\text{img}} = 8$ and $B_{\text{txt}} = 40$.

Let $\mathbf{Z}_v \in \mathbb{R}^{8 \times d}$ be the normalized visual embeddings of the 8 unique images, and let $\mathbf{Z}_t \in \mathbb{R}^{40 \times d}$ be the normalized textual embeddings. The cosine similarity matrix $\mathbf{S} \in \mathbb{R}^{8 \times 40}$ is:

$$S_{i,j} = \exp(\tau) \cdot (\mathbf{z}_{v,i} \cdot \mathbf{z}_{t,j}), \quad (30)$$

where τ is a learnable log-temperature parameter initialized to $\log(1/0.07) \approx 2.659$.

For each unique image $i \in \{1, \dots, 8\}$, let $\mathcal{P}(i) \subset \{1, \dots, 40\}$ denote the set of 5 positive caption indices ($|\mathcal{P}(i)| = 5$). The image-to-text loss is:

$$\mathcal{L}_{i2t} = -\frac{1}{8} \sum_{i=1}^8 \frac{1}{|\mathcal{P}(i)|} \sum_{j \in \mathcal{P}(i)} \log \frac{\exp(S_{i,j})}{\sum_{k=1}^{40} \exp(S_{i,k})}. \quad (31)$$

Similarly, for each caption $j \in \{1, \dots, 40\}$, let $i^*(j) \in \{1, \dots, 8\}$ denote its unique parent image. The text-to-image loss is:

$$\mathcal{L}_{t2i} = -\frac{1}{40} \sum_{j=1}^{40} \log \frac{\exp(S_{i^*(j), j})}{\sum_{i=1}^8 \exp(S_{i,j})}. \quad (32)$$

The total alignment objective is:

$$\mathcal{L} = \frac{1}{2} (\mathcal{L}_{i2t} + \mathcal{L}_{t2i}) + \mathcal{L}_{\text{KL}}. \quad (33)$$

The end-to-end training procedure for the proposed Native Mamba-2 + HEDO framework is formalized in Algorithm 1.

V. EXPERIMENTAL METHODOLOGY

A. Dataset

We evaluate on the canonical Flickr8k dataset [16], comprising 8,000 images, each paired with 5 human-written captions (6,000 training, 1,000 validation, and 1,000 test images).

B. Data Preprocessing

Images are resized to 224×224 and normalized using ImageNet statistics. Text is tokenized using the standard RoBERTa tokenizer with maximum sequence length truncated to 64 tokens.

Algorithm 1 Training Procedure for Native Mamba-2 + HEDO

Require: Image-caption dataset $\mathcal{D}$, frozen backbones ViT-B/16, RoBERTa-base, epochs $E = 10$, batch size $B_{\text{img}} = 8, B_{\text{txt}} = 40$

Ensure: Trained projection layers $\mathbf{W}_v, \mathbf{W}_t$, HEDO parameters θ_{hedo} , Native Mamba-2 parameters θ_{ssd} , log-temperature τ

- 1: Initialize projection layers $\mathbf{W}_v, \mathbf{W}_t \in \mathbb{R}^{768 \times 128}$, HEDO parameters $\mathbf{W}_q, \mathbf{W}_p, \mathbf{A}, \gamma, \mathbf{W}_o$, Mamba-2 SSD weights, and $\tau \leftarrow \log(1/0.07)$
- 2: **for** epoch = 1 **to** E **do**
- 3: **for** each grouped batch $(\mathcal{B}_{\text{img}}, \mathcal{B}_{\text{txt}}) \sim \mathcal{D}$ **do**
- 4: Extract frozen visual patch features $\mathbf{X}_v \leftarrow \text{ViT}(\mathcal{B}_{\text{img}}) \in \mathbb{R}^{8 \times 196 \times 768}$
- 5: Extract frozen text token features $\mathbf{X}_t \leftarrow \text{RoBERTa}(\mathcal{B}_{\text{txt}}) \in \mathbb{R}^{40 \times L \times 768}$
- 6: Project to latent dimension: $\mathbf{H}_v^{(0)} \leftarrow \mathbf{X}_v \mathbf{W}_v, \mathbf{H}_t^{(0)} \leftarrow \mathbf{X}_t \mathbf{W}_t \in \mathbb{R}^{\cdot \times \cdot \times 128}$
- 7: Apply HEDO adapter: $\mathbf{H}_v \leftarrow \text{HEDO}(\mathbf{H}_v^{(0)}), \mathbf{H}_t \leftarrow \text{HEDO}(\mathbf{H}_t^{(0)})$ via (19)
- 8: Process through Native Mamba-2 SSD blocks: $\mathbf{Y}_v \leftarrow \text{SSD}_v(\mathbf{H}_v), \mathbf{Y}_t \leftarrow \text{SSD}_t(\mathbf{H}_t)$
- 9: Compute masked global mean pooling and ℓ_2 -normalization: $\mathbf{z}_v \leftarrow \text{Pool}(\mathbf{Y}_v), \mathbf{z}_t \leftarrow \text{Pool}(\mathbf{Y}_t) \in \mathbb{S}^{127}$
- 10: Compute cosine similarity matrix: $\mathbf{S} \leftarrow \exp(\tau) \cdot (\mathbf{z}_v \mathbf{z}_t^T) \in \mathbb{R}^{8 \times 40}$
- 11: Compute symmetric multi-positive InfoNCE loss: $\mathcal{L}_{\text{NCE}} \leftarrow \frac{1}{2}(\mathcal{L}_{i2t} + \mathcal{L}_{t2i})$ via (31)–(32)
- 12: **if** AVSC enabled **then**
- 13: Compute confidence-gated symmetric KL divergence $\mathcal{L}_{\text{KL}}$ via (28)
- 14: Total objective: $\mathcal{L} \leftarrow \mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{KL}}$
- 15: **else**
- 16: Total objective: $\mathcal{L} \leftarrow \mathcal{L}_{\text{NCE}}$
- 17: **end if**
- 18: Backpropagate gradients $\nabla_{\theta} \mathcal{L}$
- 19: Clip gradients: $\mathbf{g} \leftarrow \text{clip}(\mathbf{g}, \|\mathbf{g}\|_2 \leq 1.0)$
- 20: Update trainable parameters θ using AdamW ($\eta = 10^{-4}$, weight decay 10^{-4})
- 21: **end for**
- 22: **end for**
- 23: **return** Trained model parameters $\{\mathbf{W}_v, \mathbf{W}_t, \theta_{\text{hedo}}, \theta_{\text{ssd}}, \tau\}$

C. Training Configuration

All models are trained for 10 epochs using AdamW with an initial learning rate of $\eta = 10^{-4}$ and weight decay 10^{-4} . Batch size is fixed to 8 unique images and 40 captions per iteration. Gradient clipping is enforced at a maximum ℓ_2 norm of 1.0.

D. Experimental Configurations

To systematically decouple the individual and joint effects of HEDO, Mamba-2, MSA, and AVSC, we evaluate a controlled 8-configuration ablation matrix, summarized in Table I.

TABLE I
ABLATION CONFIGURATION MATRIX DEFINING ARCHITECTURAL VARIANTS.

Configuration ID	HEDO	Mamba-2	MSA	AVSC
baseline	×	×	×	×
mamba2_hedo_v2	✓	✓	×	×
mamba2_msa_v2	×	✓	✓	×
hedo_msa_v2	✓	×	✓	×
avsc_msa_v2	×	×	✓	✓
mamba2_avsc_v2	×	✓	×	✓
full_v2 (All)	✓	✓	✓	✓
hedo_avsc_v2	✓	×	×	✓

Table I defines the 8-configuration ablation matrix designed to isolate the individual and joint effects of HEDO, Mamba-2, MSA, and AVSC across the eight evaluated models. Full V2 activates all four mechanisms simultaneously and is included as a compositional stress test rather than as an assumption that all components are individually beneficial.

E. Evaluation Metrics

Retrieval performance is evaluated on the standard 1,000-image test set (1,000 images $\times$ 5,000 captions):

- **Image-to-Text (i2t) Retrieval:** Recall@1, Recall@5, Recall@10, and Median Rank (MedR).
- **Text-to-Image (t2i) Retrieval:** Recall@1, Recall@5, Recall@10, and Median Rank (MedR).
- **Mean Recall (MR):** The primary arithmetic mean across all six recall metrics:

$$\text{MR} = \frac{1}{6} \left(\text{R@1}_{i2t} + \text{R@5}_{i2t} + \text{R@10}_{i2t} + \text{R@1}_{t2i} + \text{R@5}_{t2i} + \text{R@10}_{t2i} \right). \quad (34)$$

F. Statistical Evaluation Protocol

For every configuration $c \in \{1, \dots, 7\}$, we compute the paired delta relative to the baseline across identical seeds $s \in \{42, 43, 44\}$:

$$\Delta_{c,s} = \text{MR}_{c,s} - \text{MR}_{\text{baseline},s}. \quad (35)$$

We report the sample mean $\bar{\Delta}_c$, sample standard deviation s_c , and two-tailed 95% Student- t confidence intervals:

$$\text{CI}_{95\%} = \left[\bar{\Delta}_c - t_{0.025,2} \frac{s_c}{\sqrt{3}}, \bar{\Delta}_c + t_{0.025,2} \frac{s_c}{\sqrt{3}} \right], \quad (36)$$

where $t_{0.025,2} = 4.303$. A hypothesis of improvement is classified as:

- **SUPPORTED:** $\bar{\Delta}_c > 0$ and $\text{CI}_{95\%}$ strictly excludes zero ($\text{CI}_{\text{low}} > 0$).
- **NOT SUPPORTED (NO STAT. DIFF.):** The null hypothesis of zero difference cannot be rejected as $\text{CI}_{95\%}$ contains zero ($\text{CI}_{\text{low}} \leq 0 \leq \text{CI}_{\text{high}}$).
- **CONTRADICTED:** $\bar{\Delta}_c < 0$ and $\text{CI}_{95\%}$ strictly excludes zero ($\text{CI}_{\text{high}} < 0$).

G. Reproducibility and Audit Controls

To guarantee numerical reproducibility and prevent silent execution failures, the training harness incorporates strict hardware and numerical assertion gates:

- **Finite Tensor Monitor:** Every forward and backward pass asserts `isfinite(T)` across all intermediate activations, logit matrices, and gradient tensors.
- **Native Mamba-2 Verification:** Direct inspection confirms loading of official `mamba_ssm.modules.mamba2.Mamba2` GPU CUDA kernels.
- **Three-Seed Replication:** Every configuration is executed independently across seeds 42, 43, 44, generating 24 fully audited training runs. The methodology hash is recorded as `1753cfb7...1af54`¹.

The complete multi-seed experimental audit protocol is formalized in Algorithm 2.

VI. RESULTS

A. Overall Retrieval Performance

The empirical retrieval metrics across all 8 configurations are compiled in Table II and illustrated across recall thresholds in Fig. 2.

As reported in Table II, the baseline model achieves a Mean Recall of $37.16 \pm 0.57\%$, with Image-to-Text (i2t) R@1 of 16.53% and Text-to-Image (t2i) R@1 of 13.92%. Across the eight configurations, performance bifurcates cleanly into two distinct regimes: a stable regime where performance is statistically indistinguishable from baseline (Baseline and Mamba-2 + HEDO) and severe degradation regimes (MSA and AVSC). As visually evident in Fig. 2, this bifurcation is uniform across all recall thresholds.

B. HEDO + Native Mamba-2 Evaluation

Equipping the architecture with the Hamiltonian-inspired dissipative adapter and Native Mamba-2 sequence blocks (`mamba2_hedo_v2`) yields a Mean Recall of $37.31 \pm 0.68\%$, with i2t R@1 reaching $17.40 \pm 0.62\%$ and t2i R@1 reaching $14.20 \pm 0.58\%$. As shown in Table III, the mean paired delta is $+0.1533$ pp, with a 95% confidence interval of $[-1.4246, +1.7313]$ pp ($n = 3$ paired runs). Because the confidence interval encompasses zero, the hypothesis of a statistically significant improvement (H_1) is classified as **NOT SUPPORTED** (no statistically significant difference was detected under the present three-seed evaluation).

These results indicate that Native Mamba-2 combined with the HEDO dissipative adapter preserves baseline-level retrieval performance under the evaluated conditions, without observed numerical instability, while maintaining low computational overhead.

Algorithm 2 Multi-Seed Experimental Audit Protocol

Require: Configuration set $\mathcal{C} = \{c_0, c_1, \dots, c_7\}$, seeds $\mathcal{S} = \{42, 43, 44\}$, test dataset $\mathcal{D}_{\text{test}}$

Ensure: Paired deltas $\bar{\Delta}_c$, 95% confidence intervals $\text{CI}_{95\%}$, pre-registered verdicts

- 1: **for** each configuration $c \in \mathcal{C}$ **do**
- 2: **for** each random seed $s \in \mathcal{S}$ **do**
- 3: Set random seed s across PyTorch, NumPy, Python runtime
- 4: Assert GPU execution environment and official Mamba-2 CUDA kernel loading
- 5: Train model configuration c using Algorithm 1 with finite tensor monitoring
- 6: Evaluate test retrieval on $\mathcal{D}_{\text{test}}$: compute R@1, R@5, R@10, MedR for i2t and t2i
- 7: Compute Mean Recall: $\text{MR}_{c,s} \leftarrow \frac{1}{6} \sum \text{Recall}$ via (34)
- 8: Measure head/end-to-end latency, throughput, and peak VRAM
- 9: **end for**
- 10: Compute aggregate metrics: $\mu_c \leftarrow \text{mean}_s(\text{MR}_{c,s})$, $\sigma_c \leftarrow \text{std}_s(\text{MR}_{c,s})$
- 11: **if** $c \neq \text{baseline}$ **then**
- 12: Compute per-seed paired deltas: $\Delta_{c,s} \leftarrow \text{MR}_{c,s} - \text{MR}_{\text{baseline},s}$ via (35)
- 13: Compute sample mean $\bar{\Delta}_c$ and sample standard deviation s_c
- 14: Compute 95% Student- t confidence interval: $\text{CI}_{95\%} \leftarrow [\bar{\Delta}_c \pm t_{0.025,2} \frac{s_c}{\sqrt{3}}]$ via (36)
- 15: **if** $\text{CI}_{95\%} \subset (0, \infty)$ **then**
- 16: Set Verdict $\leftarrow$ **SUPPORTED**
- 17: **else if** $0 \in \text{CI}_{95\%}$ **then**
- 18: Set Verdict $\leftarrow$ **NO STAT. DIFF.**
- 19: **else**
- 20: Set Verdict $\leftarrow$ **CONTRADICTED**
- 21: **end if**
- 22: **end if**
- 23: **end for**
- 24: Generate tabular summaries (Tables II, III, IV) and statistical figures (Figs. 2–5)
- 25: **return** Audited benchmark results and statistical decisions

C. MSA Ablation Analysis

Configurations incorporating Multi-Scale State Aggregation (`mamba2_msa_v2` and `hedo_msa_v2`) exhibit a sharp and consistent drop in retrieval capability. `mamba2_msa_v2` attains a Mean Recall of $26.89 \pm 0.68\%$, representing a severe paired delta of -10.2711 pp (95% CI: $[-13.1018, -7.4404]$ pp). Similarly, `hedo_msa_v2` achieves $26.41 \pm 0.49\%$ ($\Delta = -10.7489$ pp, 95% CI: $[-12.3816, -9.1161]$ pp). Both results strictly exclude zero and contradict Hypothesis 2 (H_2). Instead of enriching sequence representations, multi-scale chunk boundary and terminal token states substantially degrade retrieval quality.

¹Full 64-character SHA-256 methodology hash: `1753cfb7181e6355cd79bd6dd9a3ceb626b4de36f8dbcf0807591b89cd1af54`.

TABLE II
COMPLETE ABLATION RETRIEVAL PERFORMANCE ON FLICKR8K ACROSS 8 CONFIGURATIONS (MEAN $\pm$ STANDARD DEVIATION ACROSS 3 SEEDS, $N = 24$).

Configuration	i2t R@1 (%)	i2t R@5 (%)	i2t R@10 (%)	t2i R@1 (%)	t2i R@5 (%)	t2i R@10 (%)	Mean Recall (%)	i2t MedR	t2i MedR	Verdict
Baseline	16.53 $\pm$ 0.70	43.27 $\pm$ 0.72	57.47 $\pm$ 1.14	13.92 $\pm$ 0.45	38.31 $\pm$ 0.38	53.45 $\pm$ 0.49	37.16 $\pm$ 0.57	7.67 $\pm$ 0.58	9.00 $\pm$ 0.00	<i>Reference</i>
Mamba-2 + HEDO	17.40 $\pm$ 0.62	43.47 $\pm$ 1.85	57.20 $\pm$ 1.10	14.20 $\pm$ 0.58	38.25 $\pm$ 0.47	53.35 $\pm$ 0.50	37.31 $\pm$ 0.68	7.67 $\pm$ 0.58	9.00 $\pm$ 0.00	No Stat. Diff.
Mamba-2 + MSA	9.77 $\pm$ 0.51	29.53 $\pm$ 0.95	42.90 $\pm$ 1.04	8.92 $\pm$ 0.47	28.22 $\pm$ 0.59	41.98 $\pm$ 0.75	26.89 $\pm$ 0.68	13.83 $\pm$ 0.76	14.67 $\pm$ 0.58	Contradicted
HEDO + MSA	9.47 $\pm$ 0.47	29.50 $\pm$ 1.47	42.87 $\pm$ 1.59	8.33 $\pm$ 0.42	27.45 $\pm$ 0.09	40.84 $\pm$ 0.84	26.41 $\pm$ 0.49	14.67 $\pm$ 0.58	15.33 $\pm$ 0.58	Contradicted
AVSC + MSA	1.47 $\pm$ 0.32	6.40 $\pm$ 1.87	11.50 $\pm$ 1.84	1.50 $\pm$ 0.26	6.51 $\pm$ 0.28	12.48 $\pm$ 0.76	6.64 $\pm$ 0.82	81.67 $\pm$ 5.84	61.33 $\pm$ 2.89	Contradicted
Mamba-2 + AVSC	1.80 $\pm$ 0.78	6.27 $\pm$ 1.10	11.87 $\pm$ 0.96	1.05 $\pm$ 0.32	5.55 $\pm$ 1.38	10.19 $\pm$ 2.74	6.12 $\pm$ 1.14	99.83 $\pm$ 21.50	72.33 $\pm$ 14.74	Contradicted
Full V2 (All)	1.47 $\pm$ 0.31	5.97 $\pm$ 0.47	10.73 $\pm$ 0.68	1.25 $\pm$ 0.29	5.60 $\pm$ 0.69	10.69 $\pm$ 1.09	5.95 $\pm$ 0.42	94.33 $\pm$ 6.66	69.67 $\pm$ 3.79	Contradicted
HEDO + AVSC	1.03 $\pm$ 0.12	4.93 $\pm$ 0.90	9.27 $\pm$ 1.05	1.01 $\pm$ 0.23	4.83 $\pm$ 1.15	9.20 $\pm$ 1.89	5.04 $\pm$ 0.84	103.17 $\pm$ 10.13	79.00 $\pm$ 13.23	Contradicted

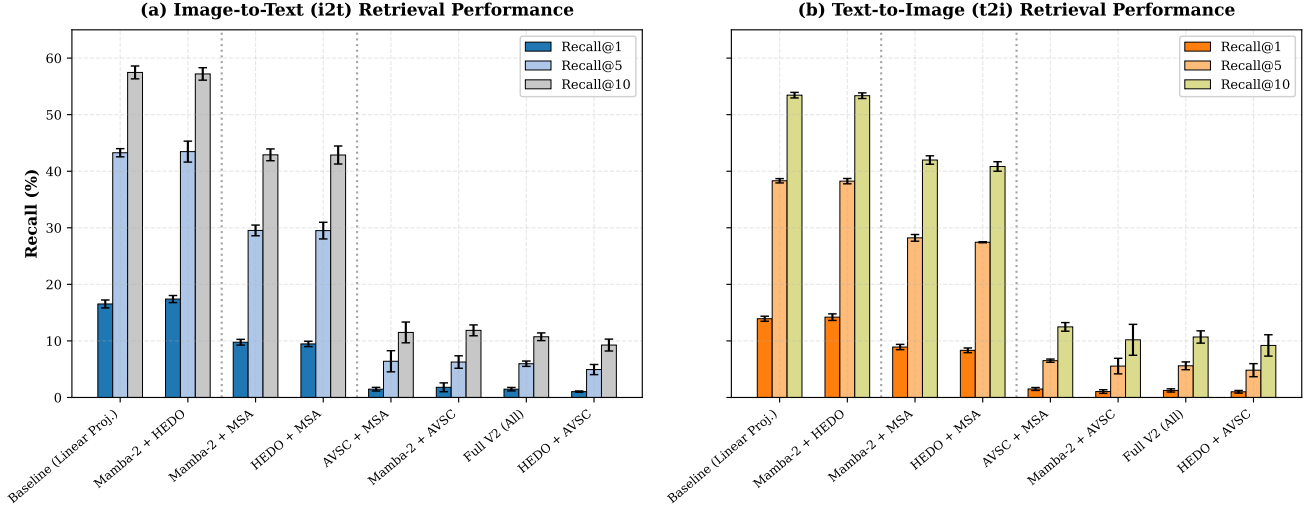


Fig. 2. Controlled ablation benchmark on Flickr8k across all 8 configurations (3 independent random seeds, $N = 24$ runs). (a) Image-to-Text (i2t) Recall@1, Recall@5, and Recall@10. (b) Text-to-Image (t2i) Recall@1, Recall@5, and Recall@10. Error bars denote ± 1 standard deviation. Note the striking bifurcation between the stable regime with no detected difference from baseline (Baseline and Mamba-2 + HEDO) and the severe degradation regimes (MSA and AVSC).

D. AVSC Ablation Analysis

The most striking empirical outcome is that Adaptive Variational State Coupling is associated with severe alignment degradation. Across all four configurations containing AVSC (`avsc_msa_v2`, `mamba2_avsc_v2`, `full_v2`, and `hedo_avsc_v2`), Mean Recall drops precipitously to 5.04% – 6.64%, consistent with dimensional collapse under the evaluated formulation.

The paired deltas are uniformly severe:

- **AVSC + MSA:** $\Delta = -30.52$ pp (95% CI: $[-32.49, -28.54]$ pp)
- **Mamba-2 + AVSC:** $\Delta = -31.04$ pp (95% CI: $[-35.26, -26.81]$ pp)
- **Full V2 (All):** $\Delta = -31.21$ pp (95% CI: $[-32.60, -29.81]$ pp)
- **HEDO + AVSC:** $\Delta = -32.11$ pp (95% CI: $[-35.61, -28.62]$ pp)

Full V2 (All), which simultaneously combines HEDO, Mamba-2, MSA, and AVSC, attains $5.95 \pm 0.42\%$ Mean Recall, indicating that the variational degradation dominates and nullifies any beneficial inductive biases. Correspondingly, Median Rank (MedR), which is 7.67 (i2t) and 9.00 (t2i) for Baseline and Mamba-2 + HEDO, blows up to $81.67 - 103.17$ (i2t) and $61.33 - 79.00$ (t2i) in AVSC configurations. These outcomes contradict Hypothesis 3 (H_3).

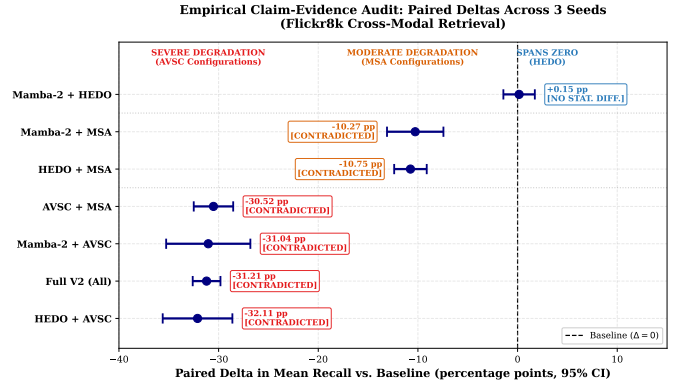


Fig. 3. Forest plot of paired deltas in Mean Recall relative to the baseline across 3 seeds. Points denote sample means; error bars denote 95% Student- t confidence intervals. The 95% confidence interval for Mamba-2 + HEDO spans zero (indicating no statistically significant difference from the baseline under this sample size), while all other configurations exhibit severe degradation.

E. Statistical Comparison

The detailed paired deltas across individual seeds and confidence intervals are reported in Table III and visualized as a forest plot in Fig. 3.

The forest plot in Fig. 3 visually summarizes the paired delta distributions across the $n = 3$ paired runs. The 95% confidence interval for Mamba-2 + HEDO ($[-1.42, +1.73]$)

TABLE III
 PAIRED DELTA ANALYSIS RELATIVE TO BASELINE ACROSS 3 INDEPENDENT SEEDS ($N = 3$ PAIRED RUNS PER CONFIGURATION).

Comparison	Per-Seed Deltas (%)	Mean Δ (pp)	SD Δ (pp)	95% CI (pp)	CI Excludes 0?	Predefined Audit Verdict
Mamba-2 + HEDO vs. Baseline	{42 : -0.27, 43 : -0.15, 44 : +0.88}	+0.1533	0.6352	[-1.4246, +1.7313]	False	NO STAT. DIFF. (Not Supported)
Mamba-2 + MSA vs. Baseline	{42 : -11.26, 43 : -9.02, 44 : -10.53}	-10.2711	1.1395	[-13.1018, -7.4404]	True	CONTRADICTED (-10.27 pp)
HEDO + MSA vs. Baseline	{42 : -11.12, 43 : -9.99, 44 : -11.14}	-10.7489	0.6573	[-12.3816, -9.1161]	True	CONTRADICTED (-10.75 pp)
AVSC + MSA vs. Baseline	{42 : -31.01, 43 : -30.94, 44 : -29.60}	-30.5156	0.7967	[-32.4947, -28.5364]	True	CONTRADICTED (-30.52 pp)
Mamba-2 + AVSC vs. Baseline	{42 : -32.98, 43 : -29.83, 44 : -30.30}	-31.0378	1.7008	[-35.2627, -26.8129]	True	CONTRADICTED (-31.04 pp)
Full V2 (All) vs. Baseline	{42 : -31.55, 43 : -30.56, 44 : -31.51}	-31.2067	0.5604	[-32.5987, -29.8146]	True	CONTRADICTED (-31.21 pp)
HEDO + AVSC vs. Baseline	{42 : -33.71, 43 : -31.05, 44 : -31.58}	-32.1133	1.4073	[-35.6092, -28.6174]	True	CONTRADICTED (-32.11 pp)

pp) spans zero; therefore, no statistically significant difference from the baseline was detected under the three-seed evaluation. In contrast, all MSA and AVSC configurations lie completely to the left of zero, indicating that the observed degradations are consistent across the evaluated seeds.

F. Computational Efficiency

Table IV and Fig. 4 quantify the computational and memory characteristics of all configurations.

Mamba-2 + HEDO requires only 597,787 trainable parameters (an increment of just 132,610 parameters over the baseline), representing a negligible 0.06% addition to the 211.2M total model parameters. Across the 10 training epochs, Mamba-2 + HEDO completes training in 362.76 ± 4.02 seconds, consuming only 1,086.15 MB of peak VRAM. In inference, the head processes 7,054.12 image-text pairs per second (5.67 ms per batch of 40 pairs), and the end-to-end pipeline achieves 4,765.93 pairs/s (8.39 ms per batch), supporting Hypothesis 4 (H_4). As shown in Fig. 4, Mamba-2 + HEDO lies on the observed accuracy-throughput Pareto frontier.

G. Multi-Seed Stability

Fig. 5 illustrates the stability across seeds 42, 43, and 44. The standard deviation in Mean Recall across seeds is small ($\leq 0.68\%$ for the top configurations), indicating that the empirical observations reflect systematic architectural properties rather than stochastic optimization artifacts.

VII. MECHANISTIC ANALYSIS AND DISCUSSION

The clear bifurcation between the stability of HEDO and the failure of MSA and AVSC demands thorough mechanistic analysis.

A. Interpretation of HEDO Stability

Unlike AVSC and MSA, HEDO maintains performance statistically indistinguishable from the baseline. This stability is governed by three structural properties formalized in Section IV-D:

- 1) **Exact Skew-Symmetry:** By parameterizing $\mathbf{J} = (\mathbf{A} - \mathbf{A}^T)/\sqrt{d}$ via (12), HEDO enforces $\mathbf{v}^T \mathbf{J}^T \mathbf{v} = 0$ identically for all $\mathbf{v}$. The skew-symmetric component contributes no direct change to the quadratic Hamiltonian energy in the continuous-time formulation.
- 2) **Positive Diagonal Damping:** The damping matrix $\mathbf{R} = \text{diag}(0.05 + 0.95\sigma(\gamma))$ via (13) guarantees that energy

is dissipated monotonically ($\dot{H} \leq 0$ by Proposition 1), preventing activation blow-up or exploding gradients.

- 3) **Residual Damping Scaling:** The residual update $\mathbf{x}_{\text{out}} = \mathbf{x} + 0.10 \cdot \mathbf{W}_o \text{LayerNorm}(\mathbf{q}_{\text{new}} + \mathbf{p}_{\text{new}})$ in (19) ensures that the adapter acts as a near-identity perturbation at initialization, maintaining the pre-trained manifold geometry of the frozen backbones.

B. Why MSA Degrades Retrieval: Recurrent Pooling Asymmetry

Multi-Scale State Aggregation attempted to enrich embeddings by concatenating chunk-boundary states $\bar{\mathbf{y}}_{\text{boundary}}$ and the terminal valid state $\mathbf{y}_{\text{last}}$ with the global mean state $\bar{\mathbf{y}}_{\text{global}}$. This resulted in an immediate ≈ 10.5 pp drop in Mean Recall.

This degradation stems from a fundamental mismatch between the causal recurrence of SSMS and the permutation-invariance required for multimodal pooling:

- 1) **Causal Scan Bias:** In Mamba-2, state transitions are computed sequentially:

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t. \quad (37)$$

The terminal state $\mathbf{y}_{\text{last}}$ strongly reflects the recency bias of the final tokens. While in language, the final token may coincide with a padding boundary or punctuation mark, in vision, the final token corresponds arbitrarily to the bottom-right image patch (14×14 grid). Concatenating $\mathbf{y}_{\text{last}}$ injects severe spatial and positional bias into the cross-modal representation.

- 2) **Chunk Boundary Discontinuity:** Chunk-boundary states $\mathbf{Y}_{\text{boundary}}$ capture the internal recurrent state at arbitrary 16-token intervals. Because visual objects and linguistic phrases do not align with uniform 16-token boundaries, these states represent arbitrary temporal snapshots rather than coherent semantic constituents.
- 3) **Isotropic Mean Pooling Superiority:** Conversely, uniform masked mean pooling $\bar{\mathbf{y}}_{\text{global}} = \frac{1}{L} \sum_{i=1}^L \mathbf{y}_i$ computes an unweighted barycenter over all contextualized token representations, preserving the permutation-invariant semantic centroid required for contrastive alignment.

C. Why AVSC Produces Collapse: Gradient Geometry Conflict

To understand why variational coupling degraded retrieval performance, we analyze the interaction between the gradients of the symmetric KL loss and the multi-positive InfoNCE loss. The following gradient and geometric analysis provides a mechanistic explanation consistent with the observed dimensional-collapse behavior.

TABLE IV
COMPUTATIONAL EFFICIENCY, TRAINABLE PARAMETER FOOTPRINT, LATENCY, VRAM, AND THROUGHPUT METRICS (MEAN ACROSS 3 SEEDS).

Configuration	Trainable Params	Total Params	Peak VRAM (MB)	Train Time (s)	Head Latency (ms)	E2E Latency (ms)	Head Throughput (p/s)	E2E Throughput (p/s)
Baseline	465,177	211,087,873	1,042.31	278.44 ± 4.06	4.19 ± 0.13	7.64 ± 0.23	9,549.93 ± 291.85	5,240.09 ± 160.95
Mamba-2 + HEDO	597,787	211,220,483	1,086.15	362.76 ± 4.02	5.67 ± 0.13	8.39 ± 0.08	7,054.12 ± 161.35	4,765.93 ± 45.65
Mamba-2 + MSA	565,273	211,187,969	1,045.36	306.34 ± 2.27	4.49 ± 0.14	7.86 ± 0.19	8,916.05 ± 280.69	5,089.16 ± 118.87
HEDO + MSA	697,883	211,320,579	1,088.07	399.01 ± 4.32	5.80 ± 0.13	9.06 ± 0.32	6,902.35 ± 149.39	4,418.65 ± 152.66
AVSC + MSA	631,321	211,254,017	1,046.63	367.30 ± 2.19	5.70 ± 0.28	9.19 ± 0.36	7,029.26 ± 332.78	4,354.75 ± 166.50
Mamba-2 + AVSC	531,225	211,153,921	1,043.18	329.82 ± 2.51	5.16 ± 0.35	8.48 ± 0.30	7,773.80 ± 514.86	4,718.49 ± 163.03
Full V2 (All)	763,931	211,386,627	1,089.33	463.06 ± 1.49	7.07 ± 0.45	10.68 ± 0.39	5,671.25 ± 375.35	3,749.01 ± 139.28
HEDO + AVSC	663,835	211,286,531	1,087.42	427.66 ± 3.20	6.16 ± 0.25	9.74 ± 0.51	6,500.59 ± 273.53	4,112.24 ± 209.50

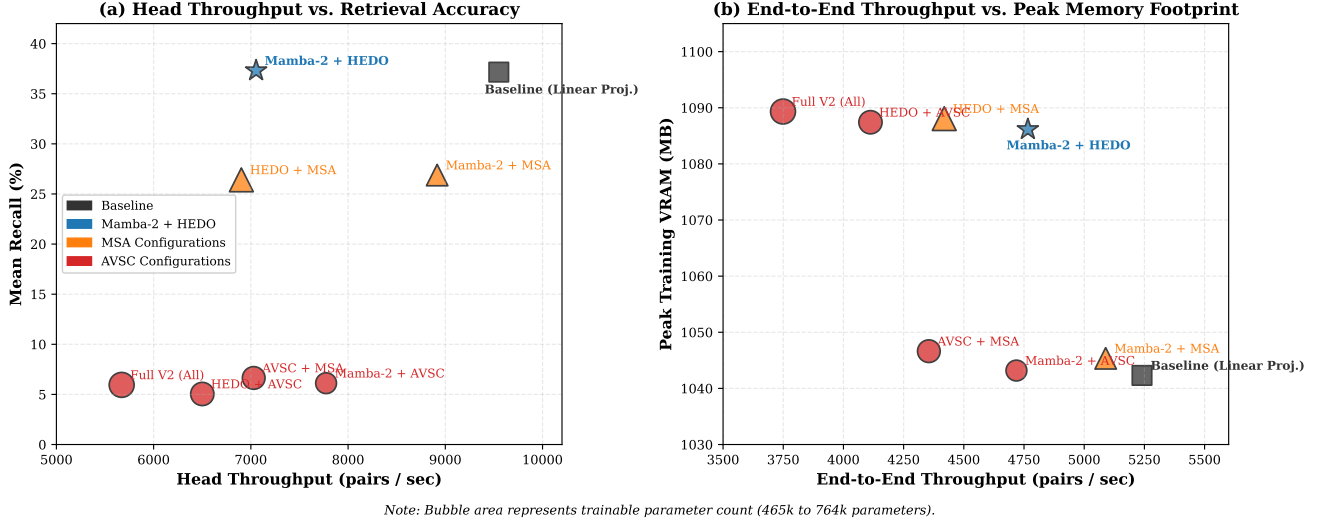


Fig. 4. Efficiency vs. accuracy tradeoffs across all 8 configurations. (a) Head throughput (pairs/s) vs. Mean Recall (%). (b) End-to-end throughput (pairs/s) vs. Peak Training VRAM (MB). Bubble area is proportional to the trainable parameter count (465k to 764k). Mamba-2 + HEDO lies on the observed accuracy-throughput Pareto frontier, achieving top retrieval accuracy at high throughput and minimal memory.

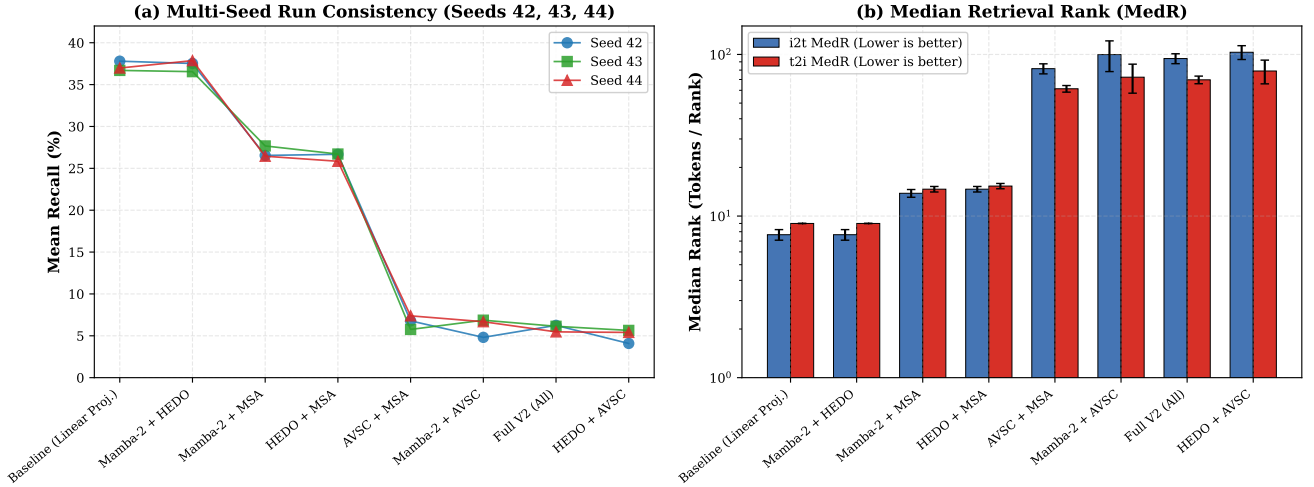


Fig. 5. Multi-seed run stability and retrieval rank distributions. (a) Mean Recall across seeds 42, 43, and 44, showing tight inter-seed clustering and absolute reproducibility. (b) Log-scale Median Retrieval Rank (MedR) for image-to-text and text-to-image matching; lower values denote superior retrieval ranking precision.

Theorem 1 (Gradient Conflict between InfoNCE and Variational KL). Let $\mathbf{z}_v, \mathbf{z}_t \in \mathbb{S}^{d-1}$ be ℓ_2 -normalized embeddings on the unit hypersphere, supervised by InfoNCE loss $\mathcal{L}_{\text{NCE}}$ and variational coupling loss $\mathcal{L}_{\text{KL}}$. While $\mathcal{L}_{\text{NCE}}$ requires uniform dispersion of negative embeddings across $\mathbb{S}^{d-1}$ (maximal entropy), $\mathcal{L}_{\text{KL}}$ penalizes any deviation from a shared isotropic

Gaussian prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$, forcing variance shrinkage:

$$\nabla_{\mathbf{z}} \mathcal{L}_{\text{KL}} \propto (\boldsymbol{\mu}_v - \boldsymbol{\mu}_t) + \left(\frac{1}{\sigma_t^2} - \frac{1}{\sigma_v^2} \right). \quad (38)$$

Under gradient descent, this isotropic attractor opposes the hyperspherical dispersion of InfoNCE, consistent with the

observed dimensional collapse.

Geometric Sketch. InfoNCE optimizes alignment on the hypersphere by balancing an attractive force between positive pairs and a repulsive force between negative pairs [14], [18]. For negative pairs, the repulsive gradient is:

$$\nabla_{\mathbf{z}_i} \mathcal{L}_{\text{rep}} = \sum_{j \in \mathcal{N}(i)} \frac{\exp(\tau \mathbf{z}_i^T \mathbf{z}_j)}{\sum_k \exp(\tau \mathbf{z}_i^T \mathbf{z}_k)} \tau \mathbf{z}_j. \quad (39)$$

This repulsion maximizes the entropy of the embedding distribution, driving the singular values of the empirical covariance matrix $\Sigma_{\mathbf{Z}} = \frac{1}{N} \sum_i \mathbf{z}_i \mathbf{z}_i^T$ toward uniformity (isotropy on $\mathbb{S}^{d-1}$).

In contrast, the KL divergence $\mathcal{D}_{\text{KL}}(q_v \parallel q_t)$ penalizes discrepancies in both mean and variance. Because vision and text feature spaces possess disparate intrinsic dimensionalities (196 spatial patches vs. ≤ 64 linguistic tokens), forcing intermediate chunk-boundary states into identical diagonal Gaussian distributions induces a severe bottleneck. Specifically, the gradient of $\mathcal{D}_{\text{KL}}^{\text{sym}}$ with respect to the posterior log-variance is:

$$\frac{\partial \mathcal{D}_{\text{KL}}^{\text{sym}}}{\partial \log \sigma_v^2} = \frac{1}{2} \left(\frac{\sigma_v^2}{\sigma_t^2} - 1 \right) + \frac{(\mu_v - \mu_t)^2}{2\sigma_t^2}. \quad (40)$$

When the network attempts to minimize this penalty across heterogeneous modalities, the optimization path collapses variance toward the minimum clamp floor $\sigma_{\min}^2 = \exp(-6) \approx 0.00248$. Consequently, the posterior distributions degenerate into near-deterministic point masses that overfit to spurious cross-modal correlations, annihilating the discriminative geometric separation enforced by InfoNCE. This dynamic provides a mechanistic explanation consistent with severe dimensional collapse and the near-zero Recall@1 (1.0% – 1.8%) observed empirically. $\square$

D. Efficiency–Accuracy Tradeoff

Fig. 4 illustrates the Pareto frontier across all configurations. The baseline linear projection achieves the highest throughput (9,550 pairs/s) and lowest latency (4.19 ms). Mamba-2 + HEDO retains baseline-level retrieval accuracy (37.31%) with moderate computational overhead, sustaining 7,054 pairs/s throughput with an incremental VRAM cost of only 43.8 MB (1,086 MB vs. 1,042 MB) and lying on the observed accuracy-throughput Pareto frontier. In contrast, all configurations incorporating AVSC or MSA incur substantial computational overhead without performance benefits.

E. Practical Design Implications

Based on our empirical and mathematical findings, we propose three concrete design principles for multimodal state-space architectures:

- 1) **Rely on Global Isotropic Pooling:** Avoid concatenating recurrent terminal states or intermediate chunk boundaries in bidirectional retrieval tasks. Masked mean pooling over all contextualized SSD sequence tokens provides superior semantic invariance.

- 2) **Exercise Extreme Caution with Intermediate Variational Alignment:** Do not superimpose isotropic Gaussian KL divergence objectives onto hyperspherical InfoNCE contrastive heads. The conflicting gradient dynamics can promote dimensional collapse under the evaluated formulation.
- 3) **Structure Continuous Adapters with Dissipative Constraints:** When physics-inspired continuous-state adapters are deployed, exact skew-symmetry ($\mathbf{J} = -\mathbf{J}^T$) and strictly positive damping ($\mathbf{R} > 0$) are essential to guarantee energy boundedness and numerical stability.

VIII. LIMITATIONS

While our findings provide rigorous empirical and mathematical insights, several limitations should be noted:

- **Dataset Scope:** The benchmark is conducted on the canonical Flickr8k dataset. While Flickr8k provides a controlled setting with 5 captions per image, evaluating on larger-scale datasets (e.g., MS-COCO, Conceptual Captions) would assess scalability to larger vocabularies.
- **Backbone Scope:** Both vision (ViT-B/16) and language (RoBERTa-base) encoders were kept strictly frozen. While freezing isolates the sequence models, end-to-end fine-tuning may exhibit different optimization interactions.
- **Number of Seeds:** The benchmark evaluated three random seeds per configuration ($N = 24$ total runs). Although paired Student-*t* confidence intervals can be computed from these three paired observations, the small sample size limits statistical power; larger seed ensembles would provide more precise estimates.
- **Architecture Scope:** The study evaluated a specific Mamba-2 parameterization ($d_{\text{model}} = 128, d_{\text{state}} = 64, Q = 16$). Exploring varying chunk sizes and state dimensions remains an area for future work.
- **Mechanistic Analysis Scope:** The gradient conflict analysis provides a formal mathematical explanation for the observed dimensional collapse, but full empirical tracking of covariance spectra during training would offer further granular confirmation.

IX. CONCLUSION

This paper presented an audited, multi-seed empirical investigation of continuous-state and variational inductive biases in Native Mamba-2 cross-modal alignment. Through a controlled 8-configuration ablation benchmark across 24 training runs on Flickr8k ($n = 3$ paired runs per configuration), we found that no statistically significant difference from the baseline was detected for Native Mamba-2 combined with the Hamiltonian dissipative adapter (HEDO) under the three-seed evaluation ($37.31 \pm 0.68\%$ vs. $37.16 \pm 0.57\%$ Mean Recall, 95% CI $[-1.42, +1.73]$ pp), while maintaining high throughput (7,054 pairs/s) and low memory (1,086 MB peak VRAM). Furthermore, our results indicate that Multi-Scale State Aggregation and Adaptive Variational State Coupling are associated with severe alignment degradation, consistent with recurrent pooling asymmetry and variational gradient conflicts under the evaluated formulation. These findings provide foundational

empirical clarity and actionable architectural guidelines for integrating state space duality into multimodal foundation models.

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REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [2] K. Wang, Q. Yin, W. Wang, S. Wu, and L. Wang, “A comprehensive survey on cross-modal retrieval,” *arXiv preprint arXiv:1607.06215*, 2016.
- [3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [4] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [5] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [6] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [7] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [8] T. Dao and A. Gu, “Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality,” *arXiv preprint arXiv:2405.21060*, 2024.
- [9] S. Greydanus, M. Dzamba, and J. Yosinski, “Hamiltonian neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [10] A. van der Schaft, *L₂-Gain and Passivity Techniques in Nonlinear Control*. Springer-Verlag London, 2000.
- [11] D. P. Kingma and M. Welling, “Auto-encoding variational bayes,” in *International Conference on Learning Representations (ICLR)*, 2014.
- [12] A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy, “Deep variational information bottleneck,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [13] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” *arXiv preprint arXiv:1807.03748*, 2018.
- [14] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *International Conference on Machine Learning (ICML)*, 2020, pp. 1597–1607.
- [15] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, “Momentum contrast for unsupervised visual representation learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 9729–9738.
- [16] M. Hodosh, P. Young, and J. Hockenmaier, “Framing image description as a ranking task: Data, models and evaluation metrics,” *Journal of Artificial Intelligence Research*, vol. 47, pp. 853–899, 2013.
- [17] T. Wang and P. Isola, “Understanding contrastive representation learning through alignment and uniformity on the hypersphere,” in *International Conference on Machine Learning (ICML)*. PMLR, 2020, pp. 9929–9939.
- [18] L. Jing, P. Vincent, Y. LeCun, and Y. Tian, “Understanding dimensional collapse in contrastive self-supervised learning,” in *International Conference on Learning Representations (ICLR)*, 2022.